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Component-attached FPC manufacturing method

Abstract

A component-attached FPC manufacturing method for manufacturing a component-attached FPC in which a circuit component is solder-bonded to an FPC, includes an FPC forming step of forming an FPC that integrally includes an FPC body, an FPC arm portion branching off from the FPC body, a component mounting portion provided on an end portion of the FPC arm portion and on which the circuit component is mounted, and a bridge connecting the component mounting portion and the FPC body, a component mounting step of mounting the circuit component on the component mounting portion via a solder, and an implementing step of heating the FPC, melting the solder, and implementing the circuit component on the component mounting portion.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a component-attached FPC manufacturing method.

BACKGROUND

(2) For example, Patent Document 1 discloses a technique to mount chip components on a flexible substrate by placing the flexible substrate on a tray, placing the chip components on the flexible substrate, and mounting the chip components on the flexible substrate by reflow deposition.

(3) One example of a technique for mounting chip components on a flexible substrate by reflow deposition is a technique of mounting chip components on the flexible substrate by reflow deposition upon placing a plate having an opening on the flexible substrate to expose the chip component through the opening and pressing the flexible substrate with the plate in order to prevent the flexible substrate from warping.

RELATED ART

Patent Documents

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

(5) However, in a conventional method of mounting chip components on a flexible substrate, as illustrated in FIGS. 11, 12, when chip components 102 are mounted on a portion exposed from an opening 152a of a plate 152, i.e., a portion 103A branching off from a flexible substrate (FPC) 103, during reflow deposition, there was a concern that the portion 103A on which the chip component 102 is mounted may warp, causing the chip component (circuit component) 102 to be mounted in an out-of-position state, i.e., resulting in a mounting failure.

(6) It is an object of the present disclosure to provide a component-attached FPC manufacturing method that allows a circuit component to be mounted at predetermined location branched off from an FPC body.

Solution to Problem

(7) According to an aspect of the present disclosure, a component-attached FPC manufacturing method for manufacturing a component-attached FPC in which a circuit component is solder-bonded to an FPC, includes an FPC forming step of forming an FPC that integrally includes an FPC body, an FPC arm portion branching off from the FPC body, a component mounting portion provided on an end portion of the FPC arm portion and on which the circuit component is mounted, and a bridge connecting the component mounting portion and the FPC body, a component mounting step of mounting the circuit component on the component mounting portion via a solder, and an implementing step of heating the FPC, melting the solder, and implementing the circuit component on the component mounting portion.

(8) According to another aspect of the present disclosure, a component-attached FPC manufacturing method for manufacturing a component-attached FPC in which a circuit component is solder-bonded to an FPC, includes an FPC forming step of forming an FPC that integrally includes an FPC body, an FPC arm portion branching off from the FPC body, a component mounting portion provided on an end portion of the FPC arm portion and on which the circuit component is mounted, and an extension portion extending from a peripheral edge of the component mounting portion, a component mounting step of mounting the circuit component on the component mounting portion via a solder, a plate mounting step of mounting a plate such that at least a portion of the extension portion is covered and the component mounting portion, and an implementing step of heating the FPC, melting the solder, and implementing the circuit component on the component mounting portion.

Advantageous Effects of the Invention

(9) According to the present disclosure, a circuit component can be mounted at predetermined location branched off from an FPC body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a plan view illustrating a component-attached FPC manufactured by a component-attached FPC manufacturing method according to a first embodiment of the present disclosure.

(2) FIG. 2 is a schematic diagram for explaining the steps of the component-attached FPC manufacturing method.

(3) FIG. 3 is a diagram for explaining the manufacturing process of the component-attached FPC, illustrating a state in which a circuit component is mounted on the FPC.

(4) FIG. 4 is a diagram illustrating a process subsequent to the step of FIG. 3, illustrating a state in which a plate has been mounted on the FPC.

(5) FIG. 5 is a diagram illustrating a process subsequent to the step of FIG. 4, illustrating a state in

which the bridge on the FPC is cut.

(6) FIG. 6 is a diagram illustrating a process subsequent to the step of FIG. 5, and is a plan view illustrating a completed component-attached FPC.

(7) FIG. 7 is a plan view illustrating a modified example of a component-attached FPC manufactured by the component-attached FPC manufacturing method.

(8) FIG. 8 is a plan view illustrating a component-attached FPC according to a second embodiment of the present disclosure.

(9) FIG. 9 is a schematic diagram for explaining the steps of the component-attached FPC manufacturing method.

(10) FIG. 10 is a diagram for explaining a manufacturing procedure for manufacturing the component-attached FPC, illustrating a state in which a plate is attached to the FPC.

(11) FIG. 11 is a diagram for explaining a problem to be solved by the present disclosure, and is a plan view illustrating a chip component-attached flexible substrate manufactured by a conventional mounting method for a flexible substrate.

(12) FIG. 12 is a cross-sectional view of the chip component-attached flexible substrate illustrated in FIG. 11.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENT

First Embodiment

(13) Hereinafter, a component-attached FPC manufacturing method **1** according to the first embodiment of the present disclosure is explained with reference to FIGS. 1 to 6. FIG. 1 is a plan view illustrating a component-attached FPC **10** manufactured by a component-attached FPC manufacturing method **1** according to a first embodiment of the present disclosure.

(14) As illustrated in FIG. 1, the component-attached FPC **10** manufactured by the component-attached FPC manufacturing method **1** includes a circuit component **2** and an FPC **3** (Flexible Printed Circuits) to which the circuit component **2** is solder-bonded. In the present embodiment, the arrows X, Y, and Z are directions perpendicular to each other. The arrows X and Y indicate the planar direction in which the FPC **3** extends, and the arrow Z indicates the direction in which the circuit component **2** is mounted relative to the FPC **3**.

(15) The circuit component **2** is solder-bonded to a component mounting portion **33** of the FPC **3** (described later) and is electrically connected to a circuit formed in the FPC **3**. In this case, the circuit component **2** is an element that performs various functions. Examples of the circuit component **2** include a connector, a capacitor, a relay, a resistor, a transistor, an IPS (Intelligent Power Switch), and an electronic control unit including a microcomputer. As illustrated in FIG. 1, the circuit component **2** includes a circuit component body (not illustrated) and multiple connection portions **22** extending from both ends of the circuit component body (in the figure, only one of the multiple connection portions **22** is illustrated, and the others are omitted). In the following, one of the multiple connection portions **22** may be referred to as a “first connection portion” and the others may be referred to as “second connection portions”.

(16) In the present embodiment, the first connection portion **22** is constituted by a conductive metal tab in a rectangular plate shape. The first connection portion **22** is disposed so that a portion of the first connection portion **22** extends from the component mounting portion **33** provided on the FPC **3**, which will be described later.

(17) As illustrated in FIGS. 1, 3, the FPC **3** integrally includes an FPC body **31** formed in a rectangular shape, an FPC arm portion **32** branching off from the FPC body **31**, a component mounting portion **33** which is provided on the end portion of the FPC arm portion **32** and on which a first connection portion **22** of a circuit component **2** is mounted, and a bridge **34** connecting the component mounting portion **33** and the FPC body **31**.

(18) As illustrated in FIG. 3, the FPC body **31** is formed in a rectangular shape. The FPC body **31** includes a pair of first edges **31a**, **31a** that form both ends of the FPC body **31** in the left-right direction X and extend in the front-rear direction Y, and a second edge **31b** that is continuous with

the pair of first edges **31a**, **31a** and extends in the left-right direction X.

(19) As illustrated in FIG. 3, the FPC arm portion **32** includes an FPC arm portion body **321** in a strip shape extending to the front side **Y1** from the second edge **31b** of the FPC body **31**, and an FPC extension portion **322** in a strip shape bent at a right angle at an end of the FPC arm portion body **321** away from the FPC body **31** and extending in the left-and-right direction X. One end **32L** of the FPC arm portion **32** is continuous with the second edge **31b** of the FPC body **31**, and the other end **32R** is continuous with a first peripheral edge **33a** of a component mounting portion **33** described below.

(20) As illustrated in FIG. 3, the component mounting portion **33** is formed in a rectangular shape. The component mounting portion **33** includes a first peripheral edge **33a** and a second peripheral edge **33b** extending in the front-and-rear direction Y, and a third peripheral edge **33c** and a fourth peripheral edge **33d** extending in the left-and-right direction X. Furthermore, the component mounting portion **33** is provided on the opposite side of the FPC extension portion **322** from the FPC arm portion body **321** in left-and-right direction X. Solder printing is applied to the upper surface of the component mounting portion **33**. In the following, solder is given a reference numeral **4**. Further, the component mounting portion **33** is configured so that the first connection portion **22** of the circuit component **2** is electrically connected to the component mounting portion **33** via the solder **4**. That is, the circuit component **2** is mounted on the component mounting portion **33**.

(21) The bridge **34** is formed in a strip shape as illustrated in FIG. 3. The bridge **34** is situated between the FPC body **31** and the component mounting portion **33** and is provided continuously with both of them. The bridge **34** extends in the front-and-rear direction Y, with one end continuing to the second edge **31b** of the FPC body **31** and the other end continuing to the third peripheral edge **33c** of the component mounting portion **33**.

(22) Next, a manufacturing apparatus **5** used in the manufacturing the component-attached FPC **10** is explained with reference to FIG. 4.

(23) As illustrated in FIGS. 3 and 4, the manufacturing apparatus **5** includes a tray **51** on which the FPC **3** is placed, a reflow furnace (not illustrated) that melts the solder **4** by heating, and a plate **52** (illustrated in FIG. 4) that covers and presses the FPC **3**.

(24) As illustrated in FIG. 4, the tray **51** is configured so that the FPC **3** and the circuit component **2** can be placed thereon. The tray **51** is formed in a flat plate shape and is configured to have an area larger than that of the component-attached FPC **10**.

(25) As illustrated in FIG. 4, the plate **52** includes a plate body **521** that is of such a size as to cover the FPC body **31** and a portion of the FPC arm portion **32** of the FPC **3**, and an opening **522** in a rectangular shape provided in the plate body **521**. The opening **522** is of such a size as to expose the component mounting portion **33**, a portion of the FPC arm portion **32** including the other end **32R**, and the first connection portion **22** of the circuit component **2**.

(26) Next, a method for manufacturing the component-attached FPC **10** using the manufacturing apparatus **5** (the component-attached FPC manufacturing method **1**) is explained with reference to FIGS. 2 to 6.

(27) First, as illustrated in FIGS. 2, 3, the FPC **3** is formed that integrally includes the FPC body **31**, the FPC arm portion **32** branching off from the FPC body **31**, the component mounting portion **33** provided on the FPC arm portion **32** and on which the circuit component **2** is mounted, and the bridge **34** connecting the component mounting portion **33** and the FPC body **31** (FPC forming step **S1**). Then, solder is printed on the upper surface of the component mounting portion **33**. Then, the FPC **3** is placed in the tray **51**.

(28) Next, as illustrated in FIG. 4, the first connection portion **22** of the circuit component **2** are mounted on the component mounting portion **33** (component mounting step **S2**). The first connection portion **22** is placed such that a portion on the rear side **Y2** is on the component mounting portion **33**, and that a portion on the front side **Y1** crosses the fourth peripheral edge **33d**

of the component mounting portion **33**. In this manner, the circuit component **2** is mounted on the component mounting portion **33** via the solder **4**.

(29) Next, as illustrated in FIG. **4**, the plate **52** is placed on the FPC **3** (plate mounting step **S3**). The plate **52** is placed such that the plate body **521** covers the FPC body **31** and a portion of the FPC arm portion **32**, and that a portion of the FPC arm portion **32** including the other end **32R**, the component mounting portion **33**, the first connection portion **22** of the circuit component **2**, and the bridge **34** are exposed from the opening **522**. In the following, as illustrated in FIG. **4**, a state in which the FPC **3** and the circuit component **2** are placed on the tray **51** and the plate **52** is placed at a predetermined position on the FPC **3** may be referred to as a “component set **6**”. If there is no risk of warping of the FPC body **31**, the plate **52** may be omitted.

(30) Next, the component set **6** is placed in the reflow furnace. Accordingly, the FPC **3** is heated, and the solder **4** is melted. In this case, the component mounting portion **33** is supported on the FPC body **31** by the bridge **34**, and warping of the component mounting portion **33** is alleviated.

(31) Thereafter, the first connection portion **22** of the circuit component **2** is electrically connected to the circuit formed in component mounting portion **33** of the FPC **3** via the solder **4**. Also, second connection portion of circuit component **2** is electrically connected to the circuit at a predetermined position. Accordingly, the circuit component **2** is mounted on the component mounting portion **33** (implementing step **S4**). In this case, warping of the component mounting portion **33** is alleviated, so that the circuit component **2** is positioned at a predetermined position on the component mounting portion **33**.

(32) Thereafter, the component set **6** is removed from the reflow furnace, and the plate **52** is removed. Then, as illustrated in FIG. **5**, the bridge **34** is cut at a boundary position **L1** between the bridge **34** and the FPC body **31** and at a boundary position **L2** between the bridge **34** and the component mounting portion **33** (bridge cutting step **S5**). In this manner, the component-attached FPC **10** is completed as illustrated in FIG. **6**.

(33) According to the embodiment explained above, the component-attached FPC manufacturing method **1** for manufacturing the component-attached FPC **10** in which the circuit component **2** is solder-bonded to the FPC **3** includes an FPC forming step **S1** of forming the FPC **3** that integrally includes the FPC body **31**, the FPC arm portion **32** branching off from the FPC body **31**, the component mounting portion **33** provided on the end of the FPC arm portion **32** and on which the circuit component **2** is mounted, and the bridge **34** connecting the component mounting portion **33** and the FPC body **31**. Accordingly, when the FPC **3** is heated, the FPC **3** is heated in the reflow furnace while the component mounting portion **33** is supported on the FPC body **31** by the bridge **34**, so that warping of the component mounting portion **33** is alleviated. As a result, the circuit component **2** can be mounted in a predetermined orientation at the predetermined position (the component mounting portion **33**) branched off from the FPC body **31**.

(34) The plate mounting step **S3** is provided to place the plate **52** such as to cover a portion of the FPC body **31** and a portion of the FPC arm portion **32** while exposing the component mounting portion **33**. As a result, warping of the FPC body **31** is alleviated.

(35) The bridge cutting step **S5** of cutting the bridge **34** is provided, and the bridge cutting step **S5** is performed after the implementing step **S4**. Accordingly, the unnecessary portion can be cut off to produce the component-attached FPC **10** that is light in weight.

(36) It should be noted that the present disclosure is not limited to the above-mentioned embodiment, and includes other configurations that can achieve the purpose of the present disclosure, and the following modified embodiments are also included in the present disclosure.

(37) In the embodiment explained above, in the component mounting step **S2**, the circuit component **2** is mounted on the component mounting portion **33** of the FPC **3**, and thereafter, in the plate mounting step **S3**, the plate **52** is placed on the FPC **3**, but the present disclosure is not limited thereto. After the plate **52** is placed on the FPC **3**, the circuit component **2** may be placed on the component mounting portion **33** of the FPC **3**. In other words, after the plate mounting step **S3**, the

component mounting step S2 may be performed.

(38) Furthermore, in the embodiment explained above, in the bridge cutting step S5, the bridge 34 is cut at the boundary position L1 between the bridge 34 and the FPC body 31 and at the boundary position L2 between the bridge 34 and the component mounting portion 33, but the present disclosure is not limited thereto. As illustrated in FIG. 7, the bridge 34 may be cut at an intermediate position L3 in the front-and-rear direction Y, or the bridge cutting step S5 may not be performed at all, so that a state in which the component mounting portion 33 and the FPC body 31 are connected, i.e., a state in which the bridge 34 is present, may be a completed state.

Alternatively, a perforation may be provided in advance at least at one of positions L1 to L3.

Second Embodiment

(39) Hereinafter, a component-attached FPC manufacturing method 1A according to the second embodiment of the present disclosure is explained with reference to FIGS. 8 to 10. FIG. 8 is a plan view illustrating a component-attached FPC 10A according to the second embodiment of the present disclosure. FIG. 9 is a schematic diagram for explaining the steps of the component-attached FPC manufacturing method 1A. FIG. 10 is a diagram for explaining a manufacturing procedure for manufacturing the component-attached FPC 10A, illustrating a state in which a plate 52 is attached to the FPC 3A. The component-attached FPC manufacturing method 1A according to the second embodiment and the component-attached FPC manufacturing method 1 according to the first embodiment differ in the configuration of the FPC 3A (3). In the following, in the second embodiment, constituent elements having the same configuration as those of the first embodiment are denoted with the same reference numerals and detailed description thereabout is omitted.

(40) As illustrated in FIG. 8, the component-attached FPC 10A manufactured by the component-attached FPC manufacturing method 1A includes the circuit component 2 and the FPC 3A (Flexible Printed Circuits) to which the circuit component 2 is solder-bonded.

(41) As illustrated in FIG. 8, the FPC 3A integrally includes an FPC body 31, an FPC arm portion 32 branching off from the FPC body 31, a component mounting portion 33 which is provided on the end portion of the FPC arm portion 32 and on which a first connection portion 22 of a circuit component 2 is mounted, and an extension portion 35 extending from the component mounting portion 33.

(42) As illustrated in FIGS. 8 and 10, the extension portion 35 is configured to extend in a direction away from the FPC arm portion 32 at the distal end portion of the second peripheral edge 33b of the component mounting portion 33 (an end portion away from the FPC body 31 in the front-and-rear direction Y). This extension portion 35 refers to the portion extending from the second peripheral edge 33b of the component mounting portion 33.

(43) Next, a method for manufacturing the component-attached FPC 10A using the manufacturing apparatus 5 (the component-attached FPC manufacturing method 1A) is explained with reference to FIGS. 8 to 10.

(44) First, as illustrated in FIGS. 8, 9, the FPC 3A is formed that integrally includes the FPC body 31, the FPC arm portion 32 branching off from the FPC body 31, the component mounting portion 33 which is provided on the end portion of the FPC arm portion 32 and on which the circuit component 2 is mounted, and the extension portion 35 extending from the component mounting portion 33 (FPC forming step S1A). Then, solder is printed on the upper surface of the component mounting portion 33.

(45) Next, the FPC 3A is placed on the tray 51. Then, the first connection portion 22 of the circuit component 2 is placed on the component mounting portion 33 (component mounting step S2). The first connection portion 22 is placed such that a portion on the rear side Y2 is on the component mounting portion 33, and that a portion on the front side Y1 crosses the fourth peripheral edge 33d of the component mounting portion 33.

(46) Next, as illustrated in FIG. 10, the plate 52 is placed on the FPC 3A (plate mounting step S3). The plate 52 is placed such that the plate body 521 covers the FPC body 31 and a portion of the

FPC arm portion **32**, and such that a portion of the FPC arm portion **32** including the other end **32R**, the component mounting portion **33**, the first connection portion **22** of the circuit component **2**, and the proximal end of the extension portion **35** are exposed from the opening **522**. In the following, the state in which the FPC **3A** and the circuit component **2** are placed on the tray **51** and the plate **52** is placed at a predetermined position on the FPC **3A** as illustrated in FIG. **10** may be referred to as “component set **16**”.

(47) Next, the component set **16** is put into the reflow furnace. Accordingly, the FPC **3A** is heated, and the solder **4** is melted. In this case, the component mounting portion **33** is pressed by the plate **52** via the extension portion **35**, and warping of the component mounting portion **33** is alleviated.

(48) Thereafter, the first connection portion **22** of the circuit component **2** is electrically connected to the circuit formed on the component mounting portion **33** of the FPC **3A** via the solder **4**. In addition, the second connection portion of the circuit component **2** is electrically connected to the circuit at a predetermined position. As a result, the circuit component **2** is mounted on the component mounting portion **33** (implementing step **S4**). In this case, warping of the component mounting portion **33** is alleviated, so that the circuit component **2** is positioned at the predetermined position of the component mounting portion **33**.

(49) Thereafter, the component set **6** is removed from the reflow furnace, and the plate **52** is removed. In this manner, the component-attached FPC **10A** is completed as illustrated in FIG. **8**.

(50) According to the embodiment explained above, the component-attached FPC manufacturing method **1A** for manufacturing the component-attached FPC **10** in which the circuit component **2** is solder-bonded to the FPC **3A** includes an FPC forming step **S1A** of forming the FPC **3A** that integrally includes the FPC body **31**, the FPC arm portion **32** branching off from the FPC body **31**, the component mounting portion **33** which is provided on the end portion of the FPC arm portion **32** and on which the circuit component **2** is mounted, and the extension portion **35** extending from the peripheral edge of the component mounting portion **33**, and a plate mounting step **S3** of placing the plate **52** such that at least a portion of the extension portion **35** is covered and that the component mounting portion **33** is exposed. As a result, when the FPC **3A** is heated, the FPC **3A** is heated in a reflow furnace while the extension portion **35** is pressed by the plate **52**, so that warping of the component mounting portion **33** is alleviated. As a result, the circuit component **2** can be mounted in a predetermined orientation at the predetermined position (the component mounting portion **33**) branched off from the FPC body **31**.

(51) In the embodiment explained above, as illustrated in FIG. **8**, the first connection portion **22** of the circuit component **2** is placed such that a portion on the rear side **Y2** is on the component mounting portion **33**, and that a portion on the front side **Y1** crosses the fourth peripheral edge **33d** of the component mounting portion **33**. The extension portion **35** is configured to extend in a direction away from the FPC arm portion **32** in the left-and-right direction **X** at the distal end portion of the second peripheral edge **33b** of the component mounting portion **33**, but the present disclosure is not limited thereto. The first connection portion **22** of the circuit component **2** may be provided such that a portion of the first connection portion **22** is mounted on the component mounting portion **33**, and the other portion of the first connection portion **22** is mounted across the second peripheral edge **33b** of the component mounting portion **33**. In that case, the extension portion may be configured to extend in a direction away from the FPC body **31** in the front-and-rear direction **Y** at the fourth peripheral edge **33d** of the component mounting portion **33**.

(52) In the embodiment explained above, plate mounting step **S3**, the plate **52** is placed such that the proximal end of the extension portion **35** is exposed from the opening **522** of the plate **52**, but the present disclosure is not limited thereto. In the plate mounting step **S3**, the plate **52** may be placed so that the entire extension portion **35** is covered by the plate **52**.

(53) Although the best configuration, method, and the like for carrying out the present invention are disclosed in the above description, the present invention is not limited thereto. That is, although the present invention has been particularly illustrated and described primarily with respect to the

particular embodiment, it is understood that, without departing from the spirit and scope of the present invention, the present invention can be modified in various ways by those skilled in the art in terms of shape, material, quantity, and other detailed configurations of the embodiment described above. Therefore, since the descriptions that limit the shape, material, and the like disclosed above are illustrative examples to facilitate understanding of the present invention, and do not limit the present invention, a description with a name of a member, from which some or all of the limitations on the shape, material, and the like are deleted, is to be included in the present invention.

LIST OF REFERENCE SIGNS

(54) **1**, **1A** component-attached FPC manufacturing method **2** circuit component **3**, **3A** FPC **10**, **10A** component-attached FPC **31** FPC body **32** FPC arm portion **33** component mounting portion **34** bridge **35** extension portion **52** plate **S1**, **S1A** FPC forming step **S2** component mounting step **S3** plate mounting step **S4** implementing step **S5** bridge cutting step

Claims

1. A component-attached FPC manufacturing method for manufacturing a component-attached FPC in which a circuit component is solder-bonded to an FPC, comprising: an FPC forming step of forming an FPC that integrally includes an FPC body, an FPC arm portion branching off from the FPC body, a component mounting portion provided on an end portion of the FPC arm portion and on which the circuit component is mounted, and a bridge connecting the component mounting portion and the FPC body; a component mounting step of mounting the circuit component on the component mounting portion via a solder; and an implementing step of heating the FPC, melting the solder, and implementing the circuit component on the component mounting portion.
 2. The component-attached FPC manufacturing method according to claim 1, further comprising: a plate mounting step of mounting a plate such that a portion of the FPC body and a portion of the FPC arm portion are covered and the component mounting portion is exposed.
 3. The component-attached FPC manufacturing method according to claim 1, further comprising: a bridge cutting step of cutting a bridge, wherein the bridge cutting step is performed after the implementing step.
 4. The component-attached FPC manufacturing method according to claim 2, further comprising: a bridge cutting step of cutting a bridge, wherein the bridge cutting step is performed after the implementing step.
 5. A component-attached FPC manufacturing method for manufacturing a component-attached FPC in which a circuit component is solder-bonded to an FPC, comprising: an FPC forming step of forming an FPC that integrally includes an FPC body, an FPC arm portion branching off from the FPC body, a component mounting portion provided on an end portion of the FPC arm portion and on which the circuit component is mounted, and an extension portion extending from a peripheral edge of the component mounting portion; a component mounting step of mounting the circuit component on the component mounting portion via a solder; a plate mounting step of mounting a plate such that at least a portion of the extension portion is covered and the component mounting portion is exposed; and an implementing step of heating the FPC, melting the solder, and implementing the circuit component on the component mounting portion.
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